

EMM107 CAT 1

1. Discuss the effect of manganese and nickel as alloying element in steel.
 2. Discuss the effect of sulphur and phosphorous in pig iron.
 3. Relate the changes in microstructure to the properties of steels as the carbon content increases from 0.02 to 2 per cent.
 4. What are the properties of aluminum and aluminum alloys that make them attractive for Engineering applications.
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EMM107 CAT 2

1. What are the properties of aluminum and aluminum alloys that make them attractive for engineering applications?
 2. Describe two ways of forming polymers.
 3. Why are fillers added to polymers?
 4. What are the major limitations to the extensive use of composite materials in engineering applications?
 5. What are the major determinants of the properties of composite materials?
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KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (MECHANICAL
ENGINEERING)

EMM 107: ENGINEERING MATERIALS

2ND SEMESTER 2020/2021

TIME: 2 HRS

INSTRUCTIONS

Answer Question One and any other Two questions

1.
 - a) Relate the changes in microstructure and properties of steels as the carbon content increases from 0.1 to 1.2 percent. (10 marks)
 - b) Describe the mechanism of corrosion of a metal by a liquid. (8 marks)
 - c) What are the limitations in the use of composite materials in engineering applications? (6 marks)
 - d) Using the case of two elements A and B that are mutually soluble in the liquid and solid states. Illustrate the manner in which a thermal-equilibrium diagram is constructed (6 marks)
2.
 - a) What are some attractive and limiting features of structural ceramics? (3 marks)
 - b) Why are enamels and glazes added to the surface of ceramics? (3 marks)
 - c) Where do maraging steels find applications? (2 marks)
 - d) Why are few forming techniques used for thermosetting polymers? (3 marks)
 - e) Why are fillers added to polymers? (4 marks)
 - f) How are thermosetting polymers formed? (5 marks)

3. a) Explain , with aid of diagrams, the effects of the following elements to a plain carbon steel.
- i) Nickel
 - ii) Manganese (15 marks)
- b) what are some of the attractive engineering properties of aluminum and aluminum alloys? (5 marks)
4. a) Describe the two ways of forming polymers. (10 marks)
- b) List six methods of controlling corrosion. (6 marks)
- c) What factors make titanium and its alloys useful in aerospace applications? (4 marks)
5. a) What are the advantages of electro-galvanising over hot-dip galvanising? Give an outline of the latter process. (10 marks)
- b) What are the major determinants of the properties of composite materials? (5 marks)
- c) What properties make superalloys find use in jet engines, rockets and nuclear applications? (5 marks)